

TARUSH CHANDRA

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PROJECTS

- Kuckoo** | *TS, Next, Redux, Tailwind, Node, Express, PostgreSQL, GraphQL, Websockets* **Live** | **[Repo1](#)**, **[Repo2](#)**
- Designed and developed a high-performance **full-stack social media platform** serving **10+ production routes** with **13 normalized data models**, ensuring a scalable and maintainable system architecture.
 - Implemented an optimized SSR architecture with efficient data-fetching pipelines, achieving a **98/100 Lighthouse score** and **1.2s LCP** in production for a real-time social feed.
 - Developed a highly secure authentication system with **JWT-based access/refresh tokens** using **httpOnly/secure cookies** preventing XSS attacks, plus role-based authorization.
 - Built a **real-time messaging system** (1-on-1 and group chats) with WebSockets, implementing **role-based admin controls** for member management, **throttled typing indicators** reducing API calls by 85%.
 - Enhanced UX** with Next.js **intercepting routes** for modal-based post viewing while maintaining SEO-friendly URLs, **optimistic updates** for likes/comments, and **shimmer UI** across 15+ components.
 - Implemented **client-side caching** using **TanStack Query** and GraphQL API, significantly reducing redundant network requests, and complemented it with a **server-side caching** layer using **Redis** that reduces database load and improved average API response times from **650 ms to 180 ms**.
 - Redesigned database queries** using Prisma ORM with complex joins and aggregations, **eliminating N+1 queries** and minimizing database round-trips.
 - Developed a **optimized media delivery** pipeline with **AWS S3 + CloudFront** integration, achieving 65% faster load times and significant reduction in bandwidth costs.
 - Built search system with **300ms debouncing** technique to eliminate redundant API calls by **90%**, paired with recommendation algorithm analyzing user connection patterns.
 - Implemented **social graph algorithms** for follower management and mutual connection detection with **O(1)** lookup complexity through **optimized Redis indexing**.
 - Built scalable commenting system with **infinite nesting**, implemented **infinite scroll** using **cursor-based pagination** for feeds with efficient lazy loading.
 - Containerized the application stack using **Docker** and deployed on **AWS EC2**, with **Nginx** as reverse proxy and SSL termination.
 - Demo Credentials:-** Email: johndoe@gmail.com | Password: Tarush@123

SKILLS

- Frontend:** Next.js, React, Redux Toolkit, JavaScript, TypeScript, HTML5, CSS3, TailwindCSS, TanStack Query
- Backend:** Node.js, Express, WebSockets, REST APIs, GraphQL, Authentication (Stateful, Stateless), ORMs (Prisma, Mongoose)
- Databases:** PostgreSQL, MongoDB, Redis
- Infrastructure and DevOps:** AWS (EC2, S3, CloudFront, Lambda, IAM), Docker, Nginx
- Testing:** Jest, React Testing Library
- Build tools:** Bundlers (Vite, Webpack, Parcel), Package Managers (npm, yarn, pnpm), Babel
- Developer Tools:** Git and GitHub, Postman, Chrome DevTools
- Core CS Fundamentals:** Data Structures and Algorithms (DSA), Operating Systems (OS), Computer Networking (CN), Database Management Systems (DBMS), Object-Oriented Programming (OOPs), System Design (Exploring)
- Key Engineering Skills:** Scalable Architecture, Performance Optimization, Clean UI/UX

CERTIFICATIONS AND ACHIEVEMENTS

- Solved **300+ problems** on Leetcode
- JavaScript (Intermediate) — HackerRank
- Frontend Developer (React) — HackerRank
- Namaste React Course — NamasteDev

EDUCATION

National Institute of Technology, Jaipur
Bachelor of Technology in Electrical Engineering

Aug. 2018 – June 2022